Col-rank = row-rank.

Thm. (17.C.1) Let $A \in M_{m\times n}(K)$. Then col-rank(A) = row-rank(A).

Remarks. 1) Because of this Thm we'll denote from now on

rank$(A) :=$ col-rank(A) = row-rank(A).

2) Suppose A' is a row-reduced echelon matrix which is row-equivalent to A. Then we have: rank(A) = # pivots in A'.

Indeed, rank(A) = row-rank(A) = row-rank(A') = # pivots in A'.

3) Recall that for $T: V \to W$ we defined rank$(T) := \dim \mathrm{Im}(T)$.

Also, if $A \in M_{m\times n}(K)$, then rank$(T_A) = \dim \mathrm{Im}(T)$ = col-rank(A).

Preparation for the proof of the Thm.

Lemma 1. (17.C.2) Let $T: V \to W$ be a linear map, and $S: W \xrightarrow{\cong} \mathcal{U}$, $L: \mathcal{P} \xrightarrow{\cong} V$ isomorphisms. Then:

1) rank$(S \circ T)$ = rank(T).

2) rank$(T \circ L)$ = rank(T).

Proof. 1) rank$(S\circ T) = \dim S(T(V)) = \dim T(V)$ = rank(T),

b.c. $S|_{T(V)}$ is an iso. between $T(V)$ and $S(T(V))$.

2) rank$(T\circ L) = \dim T(L(\mathcal{P})) = \dim T(V)$.

(b.c. $L(\mathcal{P}) = V$.) ∎

Lemma 2. (17.C.3) Let $A \in M_{n\times n}(K)$. Then A is invertible iff $T_A: K^n \to K^n$ is an iso.

Proof. "$\Rightarrow$:" Suppose A is invertible. We have $T_{A^{-1}} = (T_A)^{-1}$.

"$\Leftarrow$:" Suppose T_A is an iso. We know that $A = [T_A]_{\mathcal{E}_n}^{\mathcal{E}_n}$, and by a prev. result we obtain that A is invertible. ∎

Lemma 3. (17.c.4) Let $A, B \in M_{m\times n}(K)$ s.t. $B = PAQ$, where $P \in GL_m(K)$, $Q \in GL_n(K)$. Then:

1) col-rank(A) = col-rank(B).

2) row-rank(A) = row-rank(B).

Proof. 1) We have $T_B = T_P \circ T_A \circ T_Q$, and both T_P & T_Q are iso's. By Lemma 1 (17.c.2), rank(T_B) = rank(T_A).

But rank(T_B) = col-rank(B), rank(T_A) = col-rank(A).

$\Rightarrow$ col-rank(B) = col-rank(A).

2) We have $B^T = Q^T \cdot A^T P^T$. Since P and Q are invertible, so are also P^T & Q^T (we have seen $(P^T)^{-1} = (P^{-1})^T$ etc.)

By part 1 of the Lemma (which we've already proven), we have

$$\begin{array}{ccc} \text{col-rank}(B^T) & = & \text{col-rank}(A^T) \\ \parallel & & \parallel \\ \text{row-rank}(B) & = & \text{row-rank}(A). \end{array}$$

∎

We are now ready for:

Proof of the Thm. (17.c.1) By a prev. Prop. (17.b.7), $\exists\, P \in GL_m(K), Q \in GL_n(K)$ s.t.

$$PAQ = \left(\begin{array}{c|c} I_r & 0 \\ \hline 0 & 0 \end{array}\right).$$ Denote this mat. by B.

For B we have $\text{col-rank}(B) = \text{row-rank}(B) = r$.

By Lemma 3 (17.C.4), $\text{col-rank}(A) = \text{col-rank}(B) = r$, and $\text{row-rank}(A) = \text{row-rank}(B) = r$. $\Rightarrow$ $\text{col-rank}(A) = \text{row-rank}(A)$. ∎

Important Remark. Although $\text{col-rank}(A) = \text{row-rank}(A)$, the spaces $\text{Col}\,S(A)$ and $\text{Row}\,S(A)$ are different.

First of all if $A \in M_{m\times n}(K)$ then $\text{Col}\,S(A) \subseteq K^m$ and $\text{Row}\,S \subseteq K^n$. But even when $n=m$ (i.e. $A \in M_{n\times n}(K)$) these two spaces are different! They have the same dim, but are different.